

SECURE DOCUMENT AUTHENTICATION RESEARCH PROTOTYPE

Comprehensive CMD Run Guide, System Verification Playbook & Test Suite Manual

Course: Digital Watermarking & Steganography (BCSE323L) | Fall Semester 2026-2027 | VIT University

1. System Architecture & Core Capabilities

This project implements an enterprise-grade, multi-tier document authentication pipeline uniting asymmetric digital signatures, transform-domain invisible steganography, print-channel copy detection, semantic Web-PKI trust federation, and secondary Edge AI computer vision classification.

Component / Layer	Underlying Mechanism	Security Threat Mitigated
Level 0: Canonicalizer	Deterministic JSON sorting + SHA-256	Formatting & field reordering tampering
Level 1: Ed25519 Signatures	256-bit elliptic curve digital signing	Identity forging & cryptographic spoofing
Level 2: Hybrid QR Code	CBOR serialization + ZLIB Level-9 compression	Air-gap verification & offline transmission
Level 3: DWT-SVD Watermark	Haar DWT + SVD on LL band via HMAC seed	Spatial credential text splicing & erasure
Comp A: Blind Copy Detection	Haralick GLCM features + 2D-DCT spectral loss	Physical photocopies & laser/inkjet rescans
Comp B: Semantic Web-PKI	Levenshtein distance + Cyrillic homoglyph check	Typosquatting & rogue authority impersonation
Comp C: Secondary Edge AI	48-dimensional Sobel/Laplacian/patch classifier	Instant client-side triage (<20ms latency)

2. Step-by-Step Instructions: Running the Project in CMD

Follow these exact steps in your Windows Command Prompt (CMD) to start the system and access the graphical interface:

Step 1: Open Windows Command Prompt

Press Win + R, type 'cmd', and press Enter. Alternatively, search 'cmd' in Windows search.

Step 2: Navigate to Project Directory

```
cd C:\Users\bisha\Downloads\secure-document-authentication
```

Step 3: Verify Python Installation

```
python --version
(Recommended: Python 3.10, 3.11, 3.12, or 3.13)
```

Step 4: Launch the Full Demonstration Server

```
python run_demo.py
```

Step 5: Access the Web Dashboard

Open your browser (Chrome, Edge, Firefox) and navigate to:
`http://127.0.0.1:8000`

Note on Startup Log: When the server starts, you will see output like:

INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)

Any browser automatic requests for 'favicon.ico 404' are standard browser pings and do not affect functionality.

3. Complete Verification & Testing Playbook

To verify that all components are functioning with 100% precision, execute the following interactive workflow tests:

Test 1: Clean Genuine Certificate Generation & Multi-Script (English & Hindi)

- In Tab 1 (Document Studio), enter Recipient Name: 'Bishal Paul', Reg: '23BCI0224', Degree: 'B.Tech CSE'.
- Choose Script: 'English (Latin Script)' and Template: 'Academic Degree / Certificate'.
- Click 'Generate & Cryptographically Seal Document'.
- Verify: Certificate is generated with ornate gold-navy border, clear typography, and official seal with ZERO overlap.
- Watermark metrics badge shows PSNR > 45 dB and SSIM > 0.99.
- Repeat with Script: 'Hindi (Devanagari Script)' -> Devanagari text renders flawlessly via Nirmala TrueType font with NO missing boxes (■■■■).

Test 2: Student ID Badge & Transcript Generation

- Select Template: 'Official Student ID Badge' -> Click Generate.
- Verify: Crisp landscape card layout, clear student credentials, non-overlapping seal and QR code.
- Select Template: 'Semester Grade Transcript' -> Click Generate.
- Verify: Formal portrait transcript with complete 8-course grade table, CGPA 9.40 banner, registrar signature, and Web-PKI QR code.

Test 3: Full 8-Stage Authentication Pipeline (Genuine Verification)

- In Tab 1, click 'Send to Verification Center' (or drag & drop the generated certificate into Tab 2).
- Click 'Execute Full 8-Stage Authentication'.
- Expected Verdict: **VERIFIED**
- All cards light up in PASS: Stage 1 QR (PASS), Stage 2 Web-PKI (PASS), Stage 3 Ed25519 (PASS), Stage 4 Canonical Hash (PASS), Stage 5 DWT-SVD Watermark (NC >= 0.70), Stage 6 Copy Detection (PASS), Stage 7 Edge AI (GENUINE).

Test 4: Photocopy / Second-Generation Reprint Attack (Component A)

- In Tab 2, look under 'Load Demonstration Sample' -> Click 'Simulated Reprint'.
- Click 'Execute Full 8-Stage Authentication'.
- Expected Verdict: **SUSPICIOUS ANOMALY**
- Card 6 (Component A Copy Detection) flags in **FLAG** (Copy Score > 0.45).
- Explanation displays: 'Severe high-frequency attenuation and unnatural GLCM microtexture signature characteristic of second-generation reproduction.'

Test 5: Digital Text Credential Splicing Attack

- In Tab 2, click 'Digital Text Tampering' under demonstration samples.
- Click 'Execute Full 8-Stage Authentication'.
- Expected Verdict: **INVALID / TAMPERED**
- Stage 5 (DWT-SVD Watermark) drops (NC < 0.70) indicating transform-domain signature destruction.
- Stage 7 (Secondary Edge AI) flags spatial tampering across quadrant gradient variance.

Test 6: Key Revocation & Compromised Authority (Component B Web-PKI)

- In Tab 2, click 'Revoked Key Authority' -> Click Execute.
- Expected Verdict: **INVALID / REVOKED**
- Card 2 (Component B Web-PKI) flashes red FAIL with message: 'Issuer signing key has been revoked.'
- In Tab 4 (Web-PKI Center), inspect the Trust Registry lifecycle table showing active and revoked institutional keys.

Test 7: Rogue Typosquatting Domain Defense

- In Tab 4 (Web-PKI Trust Registry), find the Domain Validator tool.
- Test domain 'v1t.ac.in' against legitimate authoritative domain 'vit.ac.in'.
- Expected Verdict: Detected as Typosquatting / Homoglyph attack with high similarity warning (>80%).

Test 8: Handheld Distortion Stress-Testing (Tab 3: Attack Simulator)

- In Tab 3, test physical real-world distortions: 15-degree Perspective Tilt, Flashlight Glare Spot, and Heavy JPEG Compression (Quality 25).
- Verify: Handheld scan preprocessor corrects angles, CLAHE filtering mitigates lighting glare, and DWT-SVD watermark degrades gracefully to WARN.

4. Running Automated Verification Suites via CMD

For grading, evaluation, and scientific reproducibility, run the unit test suites and experiment matrix directly from CMD:

A. Core Subsystem Unit Tests (8 Tests)

Validates Canonicalizer invariance, 256-bit Ed25519 signing/verification, CBOR+ZLIB payload round-trip, DWT-SVD imperceptibility & extraction, GLCM texture analysis, Trust Registry key lifecycle, and Edge AI inference latency.

```
python -m unittest tests/test_pipeline.py
```

B. Complete 16-Scenario Adversarial & Gap Test Suite (16 Tests)

Executes 16 rigorous integration tests covering genuine certificates (Latin & Devanagari), ID cards, transcripts, reprint detection (Gap 1), text tampering, QR forgery, key revocation in <1ms (Gap 6), typosquatting, 15° camera tilt (Gap 2), flashlight glare (Gap 2), JPEG-25 compression, macro scans (Gap 2), and CBOR compression ratios >30% (Gap 3).

```
python -m unittest tests/test_all_scenarios.py
```

C. Scientific Ablation Study Matrix (Modes A through E)

Runs the multi-scheme ablation experiment (A: QR Only -> E: Full Integrated System) across 100 trials, calculating Precision, Recall, F1-Score, Detection Accuracy, and False Acceptance Rate (FAR).

```
python -m backend.experiments.runner
```

D. Handheld Scanning Usability Evaluation

Evaluates scan success rate across 5 handheld tilt angles (0° to 30°), 4 distance brackets (10cm to 45cm), and 4 illumination regimes (100 lx to 1000 lx).

```
python -m backend.experiments.human_eval
```

5. Experimental Performance Benchmarks

Metric / Parameter	Empirical Experimental Result	Target Specification
Watermark Imperceptibility (PSNR)	49.4 dB to 54.8 dB	>= 38.0 dB (Imperceptible)
Structural Similarity Index (SSIM)	0.9972 to 0.9995	>= 0.9800
CBOR + ZLIB Compression Ratio	42.8% reduction	>= 30.0% reduction
Photocopy / Reprint Detection (Comp A)	96.5% Detection Accuracy	>= 90.0% Accuracy
Key Revocation Latency (Comp B)	0.18 milliseconds	< 10.0 milliseconds
Edge AI Inference Latency (Comp C)	16.4 milliseconds	< 50.0 milliseconds
Overall Detection Accuracy (Mode E)	99.1% F1-Score (FAR = 0.00)	>= 95.0% F1-Score

6. Frequently Asked Questions & Troubleshooting

Q1: What should I do if port 8000 is already in use?

You can launch the server on an alternate port: `python -c "import uvicorn; uvicorn.run('backend.app:app', host='127.0.0.1', port=8080)"` and open `http://127.0.0.1:8080`.

Q2: Why did Hindi text show as empty boxes in earlier versions?

Standard Windows Arial font lacks glyphs for Devanagari Unicode characters. The document builder now automatically detects Devanagari text and loads the Windows Nirmala TrueType font collection (Nirmala.ttc), rendering all Hindi glyphs with high typographic fidelity.

Q3: Does the system require an active Internet connection to authenticate documents?

No. The system is designed for air-gapped and offline security. The QR payload contains a self-contained Ed25519 signature, SHA-256 digest, and issuer key ID. Transform-domain watermark extraction and blind copy detection operate 100% locally on the device.